address_gen

December 12, 2025

[1]:
```python
# Parameters
och_tile = 1
n = 6
len_nij = n * n  # 6x6 input spatial size
n_o = 4
len_onij = n_o*n_o  # 4x4 output spatial size
R = 3
len_kij = R * R  # 3x3 kernel size

rows_per_kij = och_tile * len_onij  # 2 * 16 = 32

with open("address4b4b.txt", "w") as f:
    # Write header comments (optional, but matches your style)

    for och_t in range(och_tile):
        for pos in range(len_onij):  # 0 to 15
            x = pos % n_o
            y = pos // n_o

            for k in range(len_kij):  # 0 to 8
                k_x = k % R
                k_y = k // R
                x_in = x + k_x
                y_in = y + k_y
                nij_in = y_in * n + x_in  # Calculate nij_in
                # Calculate address
                addr = k * och_tile * len_nij + och_t * len_nij + nij_in
                # addr = k * rows_per_kij + och_t * len_onij + pos
                # Ensure 11-bit binary string
                addr_bin = format(addr, '011b')
                f.write(addr_bin)
                f.write("\n")
                # f.write("     // och_t: {}, nij_o: {}, kij: {}, addr: {}\n".
 ↪format(och_t, pos, k, addr))
```

[ ]:
```python
# Parameters
och_tile = 2
n = 6
```

```python
len_nij = n * n  # 6x6 input spatial size
n_o = 4
len_onij = n_o*n_o  # 4x4 output spatial size
R = 3
len_kij = R * R  # 3x3 kernel size

rows_per_kij = och_tile * len_onij  # 2 * 16 = 32

with open("address2b4b.txt", "w") as f:
    # Write header comments (optional, but matches your style)

    for och_t in range(och_tile):
        for pos in range(len_onij):  # 0 to 15
            x = pos % n_o
            y = pos // n_o

            for k in range(len_kij):  # 0 to 8
                k_x = k % R
                k_y = k // R
                x_in = x + k_x
                y_in = y + k_y
                nij_in = y_in * n + x_in  # Calculate nij_in
                # Calculate address
                addr = k * och_tile * len_nij + och_t * len_nij + nij_in
                # addr = k * rows_per_kij + och_t * len_onij + pos
                # Ensure 11-bit binary string
                addr_bin = format(addr, '011b')
                f.write(addr_bin)
                f.write("\n")
                # f.write("     // och_t: {}, nij_o: {}, kij: {}, addr: {}\n".
 ↪format(och_t, pos, k, addr))
```

[ ]:

```python
# check the content of address2b4b.txt and address.txt, compare if they are the␣
 ↪same
```

[2]:

```python
# Read both files
with open("address4b4b.txt", "r") as file1:
    content1 = file1.read()

with open("address.txt", "r") as file2:
    content2 = file2.read()

# Compare contents
if content1 == content2:
    print("Files are identical")
else:
    print("Files are different")
```

```python
    print(f"address2b4b.txt has {len(content1.splitlines())} lines")
    print(f"address.txt has {len(content2.splitlines())} lines")
```

```
Files are identical
```

[ ]:

```python

```